Ellie Kung

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Education

Olin College of Engineering Needham, MA
Bachelor of Science in Mechanical Engineering *Aug. 2024 – May 2028*

Experience

Mechanical Engineering Intern June 2026 – August 2026
change:Water Labs *Somerville, MA*

- Redesigned a compact, water resistant, toilet cover/seat actuation mechanism, replacing a linear actuator ramp system with a servo-driven design to address insufficient force and friction losses.
- Developed Arduino code to allow for sensor based actuation of toilet cover and to automate fatigue testing, evaluating the servo for over 100 cycles.
- Integrated an off the shelf push-lock mechanism to hold a toilet cover in place, reducing reliance on continuous servo actuation.
- Developed and evaporation tested an origami inspired membrane cover used to protect the membrane against damage.

Mechanical Engineer Sep. 2026 – Present
Olin Agriculture Robotics Research Team *Needham, MA*

- Designing a 2-axis gantry attachment for an autonomous agricultural vehicle (Amiga) to position a soil probe for automated moisture mapping, with 40 in. of y-axis travel and 20 in. of z-axis stroke.
- Performing first-principles calculations to size drive components, evaluate motor torque, decide on Acme lead screw, verify rack and pinion requirements under expected loading, and verify specs when making bill of materials.

Projects

Robotic Snake | *Onshape, 3d Printing, Arduino, Origami* Oct. 2025 - Dec. 2025

- Successfully designed and coded a six degree of freedom snake that uses bio-mimicry (snake-like movements) to traverse its environment
- Manufactured a waterproof skin based on origami pattern design iteration to protect inner components
- Lead a team of 4 through 3 sprint cycles and design reviews, coordinating design, fabrication, and integration.

Pneumatic T-Shirt Arm (PTA) | *MATLAB, CAD, Mechanical Design* Oct. 2026 - Present

- Collaborating with a three person team to design a portable, pneumatic T-shirt launcher using compressed air to propel a shirt directly through the barrel, triggered by a first pump.
- Evaluating compressed air requirements and portable tank constraints to target room-scale launches while accounting for safe operation.
- Developing an inertia-based trigger mechanism through benchtop models and CAD, using MATLAB calculations to investigate the relationship between trigger rod mass, actuation requirements, and launch-system pressure.
- Conducting preliminary safety testing by measuring velocity of a manually thrown t-shirt.

Precision Machining Projects | *Manual Milling, Turning, GD&T* Sept. 2026 - Present

- Machined precision components, including a 1-2-3 block, machinist's hammer, and clamp, using manual milling and turning processes.
- Manufactured and inspected components to drawing specifications, applying GD&T principles and precision measuring techniques.

Computational Robotics | *ROS2, RVIZ, Gazebo* Sept. 2026 - Present

- Developed a finite state machine in ROS2 that took in LiDAR and bump sensor data to follow a wall and object.
- Developed a particle filter and solved the kidnapped robot problem.

Technical Skills

Fabrication: Manual Lathe/Mill, CNC Lathe, CNC Mill, Laser Cutting, 3d Printing, Woodworking, Rapid Prototyping
Languages: Python, MATLAB, C++/Arduino, JavaScript, HTML/CSS
Tools & Platforms: OnShape, SolidWorks, ROS2, Gazebo, RViz, Git, Ubuntu, Raspberry Pi, Microsoft Office